Back to the Arcades -
A Digital Exhibition

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Link to Demo Video ♥ (with some pre-evaluation bugs present!):
https://youtu.be/jCUJDmGPe18

Link to Site ♥:
https://hedonisticopportunist.github.io/back_to_the_arcades/

1. Introduction

Building on the Digital Humanities scenario proposed in Assignment 1, this project outlines a digital museum exhibition exploring the history of arcade machines and video games, designed to engage both enthusiasts and newcomers. This scenario was chosen because arcade culture represents a form of digital heritage at risk (cf. Kobayashi, 2024, p. 3), making it a suitable context for examining how DH methods can support preservation and public engagement (cf. Owens, 2014). The exhibition highlights the cultural significance of arcade gaming and uses interactive digital technologies to make this history accessible (cf. Not et al., 2019, pp. 11–12). Inspired by *Pac-Man* creator Toru Iwatani's aspiration to engage all demographics (Retro Gamer, 2016, p. 48), the design adopts an inclusive approach that mirrors the broad appeal of classic arcade titles (cf. Therrien, 2017).

Museums play a central role in preserving cultural artefacts and making them available to the public. Digital museums extend this mission by enabling wider access, allowing people of different ages, backgrounds, and technical skill levels to engage with cultural heritage regardless of physical location (cf. Ergin, p. 130). Because digital-heritage frameworks emphasise accessibility and participatory engagement, interactive tools such as VR, mini-games, and web-based media offer an appropriate technological response to the challenges posed by the disappearance of arcade spaces (Giaccardi, 2014).

The core problem addressed here is the declining accessibility of arcade culture, particularly following the closure of many physical arcades in regions such as Japan (cf. Kobayashi, 2024, p. 3). As these spaces disappear, younger audiences lose opportunities to experience the social and historical dimensions of early video games. This project responds by developing a digital exhibition that combines VR, mini-games, and interactive media to recreate aspects of arcade environments and ensure the legacy of arcade gaming remains discoverable and inviting.

The remainder of this report outlines the intended users and requirements, describes the prototype, and presents an evaluation of its usability.

2. Users and Requirements

Design Challenge

Arcade centres across Japan—once-popular social venues—brought people from diverse backgrounds together. However, widespread closures following Covid-19 now threaten the preservation of coin-operated machines that rose to prominence in the 1970s and 1980s through titles such as Pac-Man and Space Invaders (Kobayashi, 2024, p. 3). To preserve the charm of gaming outside the home and allow younger generations to experience the social dimension of arcades (cf. Kobayashi, 2024, p. 8), this project aims to recreate arcade experiences through VR, interactive media, and accessible web-based tools.

This section outlines the intended users, their needs, and the functional and non-functional requirements derived from the scenario.

Audience

The exhibition is designed to appeal to a broad audience, reflecting the long-standing accessibility of arcade games, which are "easy to learn but hard to master" (Therrien, 2017). Different user groups benefit in distinct ways:

- Seasoned players revisit familiar classics.
- Casual visitors and newcomers engage with intuitive, low-barrier interactions.
- Nostalgic adults reconnect with games from their past.
- Families find activities suitable for various ages.

The design prioritises user experience to ensure that all visitors feel welcome, combining humanistic insights with interactive media tools such as VR (Silva et al., 2003, pp. 1–2). As Retro Gamer (2016, p. 48) notes, *Pac-Man* was intentionally designed to appeal to women and couples—an inclusive design philosophy that aligns with this exhibition's aims.

User Interface Challenges

To support this diverse audience, several UI challenges must be addressed:

- Clarity for first-time VR users, including simple movement instructions.
- Consistent navigation between the VR environment, mini-games, and interactive content.
- Accessibility for non-VR users, ensuring all content is available without specialised hardware.
- Readability and usability, especially for users with limited technical experience.
- Performance constraints, such as ensuring browser-based VR loads smoothly.

These challenges inform both the functional and non-functional requirements.

Detailed User Story

Megumi, an 18-year-old casual gamer, represents a typical visitor. She has played *Pac-Man* on emulators but has never experienced arcade culture firsthand. Using a standard laptop at home, she explores the exhibition through:

- a vintage-inspired interface,
- a branching interactive video,
- a browser-based mini-game, and
- a VR arcade environment with clear on-screen instructions.

Her journey highlights the needs of newcomers, casual gamers, and culturally curious visitors, making her an effective representative user for deriving system requirements.

Functional Requirements

Derived from Megumi's scenario using a scenario-driven design approach (Wirfs-Brock, 1993; Benyon, 2011), the functional requirements ensure that system behaviour aligns with realistic user goals.

ID	Title	Summary
UC1	Explore the VR Arcade	The visitor enters the VR arcade and moves around the virtual space.
UC2	Watch the Interactive Video	The visitor watches the interactive video on the website.
UC3	Play the *Pac-Man*-Style Game	The visitor plays the mini-game on the website.
UC4	Use the Website Hub	The visitor uses the website to access VR, the mini-game, and the interactive video.

The functional requirements below were derived directly from Megumi's scenario, using a scenario-driven design approach in which user narratives are translated into concrete system actions, as shown in the table below. This method ensures that each functional requirement corresponds to a specific user goal or interaction, grounding the system's behaviour in realistic user needs (Wirfs-Brock, 1993, p. 1).

By mapping Megumi's actions — such as navigating the VR arcade, launching the mini-game, or accessing the interactive video — to system capabilities, the design remains tightly aligned with user expectations and supports task-oriented interaction (Benyon, 2011, pp. 147–150).

ID	Requirement	Linked Use Case(s)
FR1	The system must allow visitors to enter and navigate the VR arcade.	UC1
FR2	The system must support basic VR movement (look around, walk).	UC1
FR3	The browser must allow visitors to launch and play the *Pac-Man*-style mini-game.	UC3
FR4	The interactive video must be accessible directly from the website.	UC2, UC4
FR5	The website must act as a central hub linking VR, the game, and the video.	UC4
FR6	The system must provide clear navigation between all components.	UC4
FR7	Users must be able to access the mini-game and video without entering VR.	UC2, UC3, UC4
FR8	The system must provide alternative interaction modes for users who cannot use VR.	UC4

Non- Functional Requirements

Non-functional requirements focus on qualities shaping Megumi's overall experience, including usability, accessibility, performance, and cultural-heritage alignment (Benyon, 2011; Pirrone et al., 2022).

ID	Title	Summary
S1	First-Time VR Visitor	A newcomer enters the VR arcade for the first time, explores the virtual space, and experiences the atmosphere of early arcade culture.
S2	Retro Gamer Playing the Mini-Game	An experienced gamer visits the website, launches the *Pac-Man*-style mini-game, and enjoys familiar arcade-style mechanics.
S3	Student Using the Interactive Video	A student researching digital culture watches the branching interactive video on the website and optionally explores the VR arcade for additional context.
S4	Casual Website Visitor	A visitor browses the website, watches the interactive video, tries the mini-game if interested, and may choose to enter the VR arcade.
S5	Accessibility-Focused Visitor	A visitor who avoids VR engages directly with the mini-game and interactive video on the website.

Megumi's scenario similarly informed the non-functional requirements, focusing on qualities that shape her overall experience rather than specific actions. These requirements address usability, accessibility, performance, and cultural-heritage alignment — characteristics essential for ensuring that the system is not only functional but also effective, inclusive, and contextually appropriate (Benyon, 2011, p. 149).

By analysing Megumi's needs as a first-time VR user, a casual gamer, and a culturally curious visitor, the non-functional requirements emphasise clarity, stability, responsiveness, and accessibility, reflecting broader digital-heritage principles that prioritise equitable and meaningful engagement (Pirrone et al., 2022, p. 1). These qualities also form the basis for evaluating the prototype's usability and accessibility.

ID	Requirement	Related Scenario(s)
NFR1	The interface must be intuitive for first-time visitors.	S1
NFR2	Instructions for VR movement must be clear.	S1
NFR3	Interactions across VR and the website must be simple and consistent.	S1, S2
NFR4	The VR environment must load within a reasonable time.	S1
NFR5	The mini-game must run smoothly in a browser.	S2
NFR6	The interactive video must stream without buffering under normal conditions.	S3
NFR7	The VR environment must run on standard desktop browsers.	S1
NFR8	The website must be responsive on both desktop and mobile devices.	S4
NFR9	The system must provide non-VR alternatives for all core content.	S5
NFR10	Text-based information must meet readability standards.	S3, S5
NFR11	Audio elements must include captions or text equivalents.	S3, S5
NFR12	Navigation between VR, the mini-game, and the video must be stable and crash-free.	S4
NFR13	The VR environment must remain stable during typical sessions.	S1
NFR14	The VR environment should evoke the atmosphere of a classic arcade.	S1, S2

Usage of Interactive Technologies

The exhibition incorporates multiple interactive elements to engage visitors with different interests and abilities:

- Micro-versions of classic games (e.g., *Pac-Man*, *Space Invaders*) allow users to experience the tactile legacy of arcade gaming, lowering barriers to engagement (Sayers et al., 2016, pp. 1–3).
- Audio-capture tools enable visitors to record or listen to stories from former players, adding authentic perspectives (Not et al., 2019, pp. 11–12).
- VR elements immerse users in a simulated arcade environment, enhancing emotional involvement and supporting personalised experiences (Capuano et al., 2016, pp. 968–969; Not & Petrelli, 2019, pp. 68–69).

These technologies reflect broader digital-heritage goals of democratising access and supporting participatory engagement (Pirrone et al., 2022, p. 1; Parry, 2007, pp. 65–67).

3. Prototype

High-Level Overview

The prototype is a frontend site with interactive elements, including links to mini-games, a VR environment, and an interactive experience. More specifically, the frontend (Figure 1) was built with HTML, JavaScript, and CSS, using the Pico framework to make the HTML responsive (Pico CSS, n.d.). Moreover, Pico offers the advantage of being both performance-friendly and simplifying the styling of HTML elements (mfyz, 2026). Design-wise, the site opts for a deliberately vintage look, choosing simplicity over flashiness to evoke nostalgia in users while also being user-friendly and accessible, with Pico designed for that purpose (cf. Pico CSS, n.d.).

The VR content, written in AFrame, was purposely kept simple and browser-based to avoid the cost of a) the user requiring a headset and b) avoiding issues usually associated with VR, such as motion sickness (cf. Hein et al., 2019, p.1). Browser-based VR was also selected for performance reasons, since lightweight immersive systems reduce hardware demands and broaden accessibility (Capuano et al., 2016, pp. 968–969). The VR environment — as shown in Figure 2 — contains a retro-inspired landscape in soft colours, two arcade machines and a sun/planet combination.

The mini game (Figure 3) – exclusive to this project engages p5.js, a lightweight JavaScript library that allows the content to be embedded within an HTML page (Processing Foundation, n.d.). The other games, save *Hungry Space Cat*, were also developed using p5.js. H*ungry Space Cat* is a feline-inspired tribute to arcade games with various levels; however, it was added to the site because of its arcade-heavy inspiration.

The interactive content (Figure 4) was made in Lumi and displays some information about arcade games.

Components

To provide a clearer overview of how each part of the system supports user needs, accessibility, and performance requirements, the table below integrates both accessibility considerations and known limitations for each component.

Component	Technologies Used	Design Rationale	Accessibility	Limitations
Frontend with links to: • VR environment • Mini-games • Interactive slideshow	JavaScript for dynamic elements; Pico CSS for lightweight styling.	Provides a simple, intuitive entry point for navigating all system components.	Semantic HTML tags and ARIA attributes can support screen readers and keyboard navigation (W3C, 2018).	Accessibility depends on correct implementation; Pico CSS offers limited customisation.
VR environment allowing user movement.	AFrame is a framework that allows VR to be incorporated into HTML (AFrame).	Creates an immersive arcade-themed environment to enhance engagement.	VR can be adapted for users with visual impairments by adjusting contrast and text size and by offering multiple input methods (Sarangam, 2022).	VR may still be difficult for users with motion sensitivity or limited hardware.
An assortment of mini-games inspired by arcade titles.	One custom mini-game was built with p5.js; others were created in previous projects using Unity or p5.js.	Designed to capture the “easy to start, hard to master” feel of classic arcade games.	Simple controls and visual clarity can support accessibility when intentionally implemented (Unity Technologies, 2023).	Accessibility varies across games; older projects may not follow consistent standards.
Interactive “video” experience.	Lumi for branching scenarios using videos, images, and text (Lumi Education, n.d.).	Offers an engaging, choice-driven way for users to absorb information.	Can be made accessible through audio narration, benefiting visually impaired users (Lumi Education, n.d.)	Users with slow or unstable internet connections may experience loading issues.

4. Evaluation

Types of Evaluation

There are two types of evaluation: one involves a usability expert reviewing the design, while the other has people play with a prototype. Because the latter method more closely aligns with capturing end users' needs, participation-based evaluation methods are preferred (Benyon, 2011, p. 226).

Participants

In this project, testing was conducted by a single participant — the developer — due to budget limitations; however, this was mitigated by the tester's professional background as a software tester, ensuring the evaluation remained systematic and informed despite the limited sample. The participant used a gaming laptop. For the open-ended questionnaire (Figure 7), the idea was to gather responses from several people with diverse backgrounds and ages, with two participants.

Although the number of participants was limited, the testing approach remains valid because exploratory evaluation methods prioritise depth of insight over sample size, particularly in early-stage prototypes. As Benyon (2011, p. 226) notes, participation-based evaluation is effective even with small groups when the goal is to uncover usability issues and understand how users interact with core features. Furthermore, the primary tester’s professional background in software testing strengthened the reliability of the findings, as experienced testers can identify patterns, edge cases, and inconsistencies that novice users may overlook (Kumari et al., 2018, pp. 323–332). While broader testing would enhance generalisability, the combination of expert-informed exploratory testing and targeted user feedback provides a meaningful foundation for evaluating the prototype at this stage.

Methods

Exploratory Manual Testing

Drawing on the requirements in Section 2 as test cases, exploratory manual testing is a process that, without using software tools, checks whether a system’s functionality performs as expected (Thant & Tin, 2023, p. 88). As the name suggests, exploratory testing takes an unscripted approach, focusing on software usability and functionality (GeeksforGeeks, 2025).

Automated Testing

While manual testing is crucial to software development, it also has drawbacks, such as taking up a lot of time and, because it is repetitive, can bore the tester, leading them to miss obvious bugs (Kumari et al., 2018, pp. 323-333). To spot glaring issues quickly — especially those introduced during bug fixes — a separate Cypress test application was created, enabling quick identification of such issues. Cypress was specifically chosen for this project because of its ease of installation and browser-based execution (Cypress, 2025).

Open-Ended Questions

Participants interested in trying out the game were asked the following:

- Do you find the site's layout intuitive?
- Does the site's layout remind you of arcade games?
- What did you think of the VR experience?
- What did you think of the games?
- What did you think of the interactive experience?

Manual Testing Results

Some bugs were discovered during the manual testing phase, with some even present in the demo video:

ID	Test Steps	Results	Action	Requirements Addressed	Explanation
1	Navigate through the website, with an emphasis on the mini games section, meaning: - Navigate to the mini games section of the site. - Click on every single button. - Once there, play a game and then return to the main page.	During this stage, it was discovered that the mini game specifically created for this project lacked a button to return the user to the main page. There was also no warning on the Mini Games page that the other games were hosted on external sites.	Both issues were addressed by: - Creating a button on the game's page that navigates the user back to the main page. - Adding a small note on the Mini Games page that the two other buttons lead to external sites.	FR6 – Clear navigation between all components	Both issues relate directly to navigation clarity. Users must be able to move between components without confusion, and external links must be clearly communicated to avoid disorientation.
2	Go through the interactive web experience, more specifically: - Verify that the text elements on the page make sense. - Click on the link leading to the experience.	It was discovered that the site's text elements (e.g., the header) were misleading, as the Lumi experience is less of a video experience and more of an interactive one.	The link to the page itself ("/video") was not changed, but the header was renamed from "Interactive Video Experience" to "Interactive Experience".	FR4 – Interactive video must be accessible from the website	The feature worked, but the label created a mismatch between user expectations and system behaviour. Correcting the header ensures the feature is accurately represented and easy to identify.
3	Play the mini-game, more specifically: - Navigate to the "Mini-Games" section of the page. - Play the game. - Try to play it again.	While creating the video, it became apparent that there was no reload page button on the mini-game page.	A Reload button was added.	NFR5 – Mini-game must run smoothly in a browser	Smooth gameplay includes the ability to restart without friction. Adding a reload button supports usability and ensures the game behaves as expected during repeated play.
4	Repeat steps in #ID 3.	While a reload button was now present, the one that allowed the end-user to navigate to the game's site was rather large (Figure 5).	The buttons were resized.	NFR5 – Mini-game must run smoothly in a browser NFR8 – Responsive design	The oversized button affected the layout and usability. Resizing it improved responsiveness and ensured the interface behaved consistently across devices.

Automated Testing Results

Running the automated tests (Figure 6) provided quick feedback on whether bug fixes caused issues.

Open-Ended Question Results

The survey results are as follows: the prototype's positive aspects include the layout, VR experience, and the games themselves. Some of the issues are still present in the video, as the demo was made before the evaluation.

ID	Feedback	Results	Related Requirement(s)	Explanation
1	The main button should instead link to the games page, as this navigation is more intuitive.	The button was changed to link to the games page instead.	NFR1 – Intuitive interface NFR3 – Simple and consistent interactions	This feedback highlights a clarity issue with the navigation. Users expect the primary button to lead to the most interactive content (games), so adjusting the link improves intuitiveness and consistency.
2	The text and background lack sufficient contrast, making it a bit hard to read.	The CSS was updated to improve readability.	NFR10 – Readability standards	This directly relates to readability and accessibility. Improving contrast ensures compliance with WCAG-aligned readability expectations.
3	The VR looked good, but the user was not sure what it added.	Nothing was done here, since the text in the "VR Arcade" section describes its purpose.	NFR1 – Intuitive interface NFR2 – Clear VR instructions NFR14 – Arcade atmosphere	The VR environment visually succeeds (NFR14), but its purpose was unclear, which implies a need for clearer onboarding or contextual cues (NFR1, NFR2).
4	The interactive experience isn't very interactive; it's just a slideshow.	This is something to keep in mind for a future update, but – at the moment – this is not a high-priority issue.	FR4 – Interactive video accessible NFR9 – Non-VR alternatives for core content	The feature works functionally (FR4), but the feedback suggests users expect more interactivity. This does not violate requirements but indicates a future improvement area.
5	The game where the cat is being chased requires too much scrolling.	The game's size was minified.	NFR5 – Mini-game must run smoothly NFR8 – Responsive design	Excessive scrolling indicates a layout/responsiveness issue. Adjusting the game size improves usability and responsiveness.

Limitations of Evaluation

The evaluation process was constrained by the small number of participants, which limits the generalisability of the findings, a common issue in small-scale usability studies (Nielsen, 1994; Sauro & Lewis, 2016). While the primary tester's professional background ensured systematic identification of functional issues, this expertise may also reduce the likelihood of uncovering novice-user errors or misunderstandings, reflecting known differences between expert and novice evaluators (Nielsen & Molich, 1990; Kushniruk & Patel, 2004).

Similarly, the two questionnaire responses provide only a narrow view of user experience, particularly given the diverse audience the exhibition aims to reach, underscoring the importance of broader, more representative sampling in HCI research (Lazar et al., 2017). Despite these limitations, the evaluation still produced actionable insights that directly informed improvements to navigation, readability, and game accessibility. For a future iteration, a broader participant pool—including users with varying levels of technical familiarity—would provide a more representative understanding of usability challenges and better support the refinement of both VR and non-VR components, aligning with best practices in user-centred and VR-specific design (Preece et al., 2019; Jerald, 2016).

5. Discussion / Conclusion

As a whole, the project ran quite smoothly, and in its current shape, it seems to have fulfilled the main criteria, which are:

- The user can play mini-games that evoke the feel of famous arcade titles.
- The user can move around a VR environment, reminiscent of arcade venues.
- Through an interactive experience, the user can learn about arcade machines and games.

Overall, the evaluation results indicate that the exhibition's core components function as intended and support the project's goals. The mini-games successfully evoke the feel of classic arcade titles, fulfilling the aim of offering accessible, nostalgic interactions. The VR environment also performed reliably during testing, demonstrating that browser-based VR can provide an immersive experience without requiring specialised hardware. This aligns with several non-functional requirements, including NFR7 (VR must run on standard desktop browsers) and NFR14 (evoking a classic arcade atmosphere). The website's hub structure proved effective as well, allowing users to move between the VR space, mini-games, and interactive content in a way that supports FR5 (a central hub linking all components). These strengths suggest that the prototype provides a solid foundation for a digital exhibition that balances accessibility, interactivity, and cultural heritage goals.

The issues uncovered during testing primarily related to navigation clarity, readability, and interface consistency. These findings highlight the importance of NFR1 (intuitive interface) and NFR3 (simple, consistent interactions), as even minor inconsistencies can disrupt user flow. The need to add a return button, resize navigation elements, and improve text contrast demonstrates how usability challenges often emerge only once users interact with the system directly. The feedback on the VR environment's purpose also suggests that some users may need clearer contextual cues, even if the feature itself works correctly. These issues do not undermine the prototype's core functionality but indicate areas where refinement would enhance the overall user experience.

The project's outcomes also reflect broader digital-heritage principles. By combining VR, mini-games, and interactive media, the prototype supports participatory engagement and lowers barriers to cultural access — both central themes in digital-heritage scholarship. The ability for users to explore arcade-inspired environments and interact with retro-style games aligns with preserving and communicating cultural practices that are becoming less accessible in physical form. Even with its limitations, the prototype demonstrates how digital tools can recreate aspects of social and historical experiences associated with arcade culture, supporting the preservation of intangible heritage (cf. Parry, 2007; Not & Petrelli, 2019; Capuano et al., 2016; Pirrone et al., 2022).

While the current prototype meets the project's main criteria, the evaluation results point to several directions for future development. Increasing the number and diversity of participants would provide a more representative understanding of usability challenges, particularly for users unfamiliar with VR or retro-style games. Enhancing the interactive experience to offer more meaningful choices could also address feedback about its limited interactivity. Additional improvements to navigation clarity and onboarding — especially for the VR component — would further support first-time visitors. These refinements would strengthen the prototype's ability to serve as an engaging and accessible digital exhibition.

6. Appendix

Figures

The figures referenced throughout the report are available below.

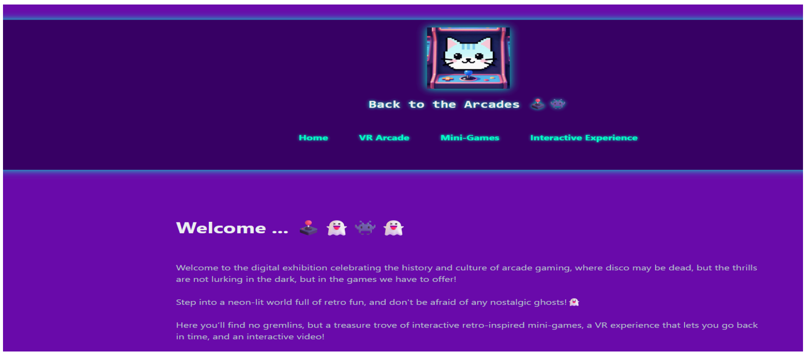


Figure 1

The frontend uses purple, retro-inspired colours.

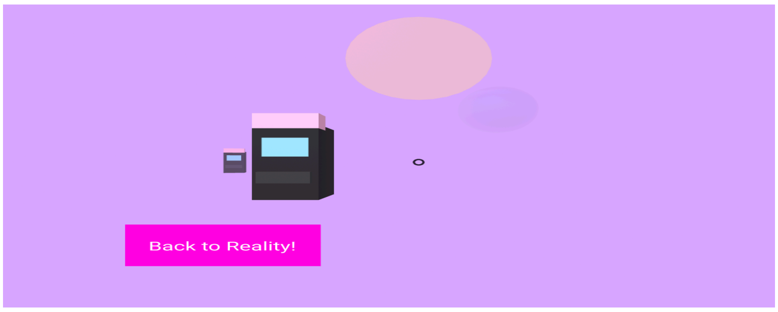
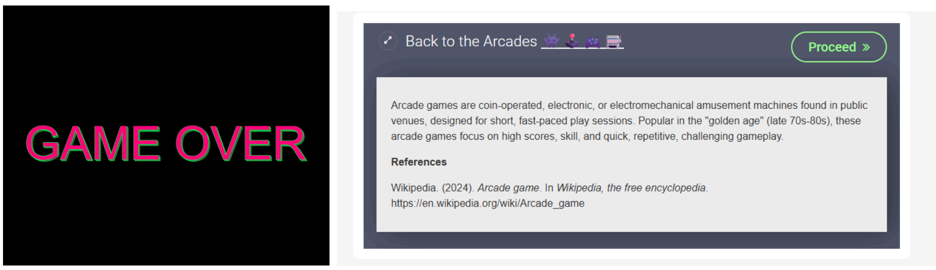


Figure 2

The VR also uses retro-inspired colours.

Note. Screenshot.



Figures 3 and 4

The mini-game and interactive experience enhance the understanding of arcade games.

Note. Screenshots.

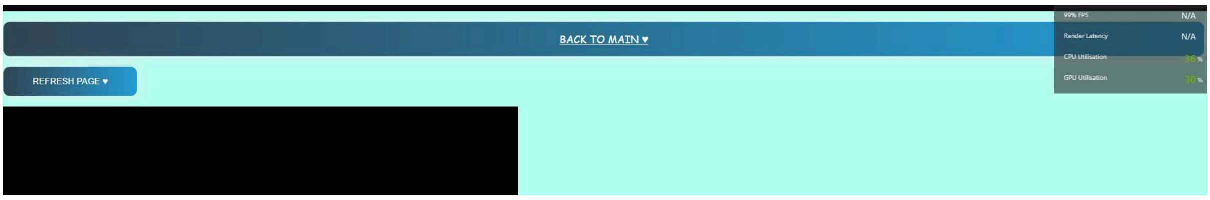
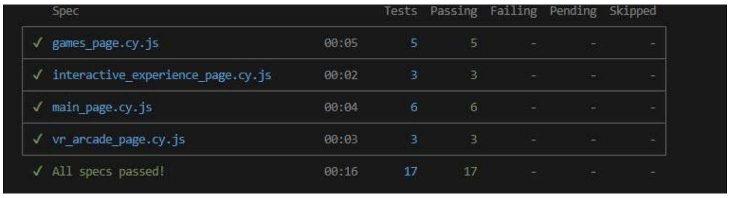


Figure 5

One of the buttons is too large.

Note. Screenshot.



Spec		Tests	Passing	Failing	Pending	Skipped
✓ games_page.cy.js	00:05	5	5	-	-	-
✓ interactive_experience_page.cy.js	00:02	3	3	-	-	-
✓ main_page.cy.js	00:04	6	6	-	-	-
✓ vr_arcade_page.cy.js	00:03	3	3	-	-	-
✓ All specs passed!	00:16	17	17	-	-	-

Figure 6

The automated test results.

Note. Screenshot.

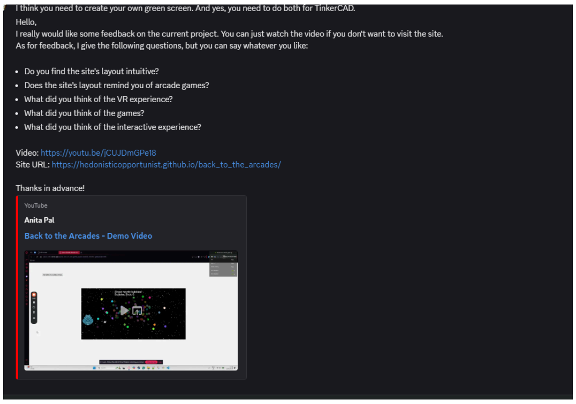


Figure 7
A screenshot of the evaluation questions.
Note. Screenshot.

Site Deployment

The frontend, with all the related interactive experiences, can be found here.

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